Numerical Methods

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

1. Why numerical methods exist
2. Root finding: bisection and Newton’s method
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1. Why numerical methods exist

Many astronomical equations cannot be solved neatly with algebra. Kepler’s equation, orbital fitting, radiative transfer and light-curve modelling often require iteration. A numerical method is a controlled recipe for producing a better answer step by step. The goal is not merely to get a number; it is to know how accurate the number is and whether the method is stable.

Three questions should always be asked: Does the method converge? How fast? What is the residual or error estimate?

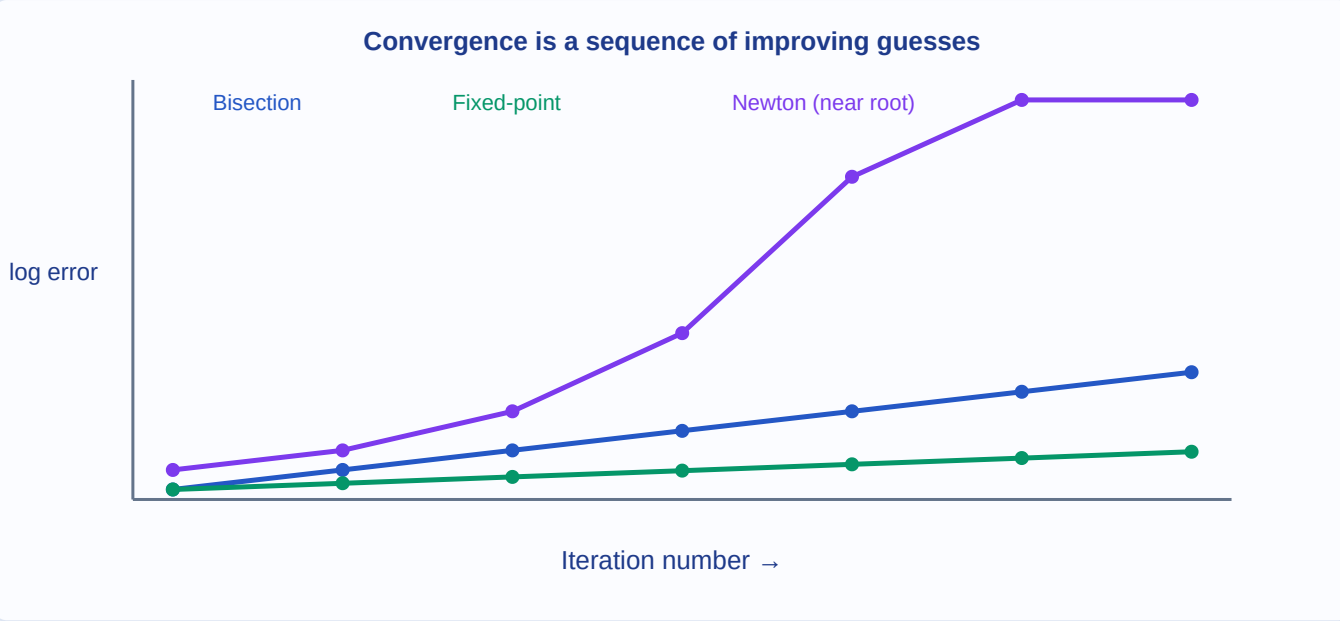


Figure 1. Schematic convergence histories. Newton's method is fast near a root; bisection is slower but robust when a sign change is bracketed.

2. Root finding: bisection and Newton's method

To solve $f(x) = 0$, bisection starts with an interval where f changes sign. Each step halves the interval. It is slow but robust. After n steps, the interval width is $(b - a)/2^n$.

Newton's method uses the tangent line:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \tag{1}$$

It can converge very quickly near a simple root, but it can fail if the derivative is small, the initial guess is poor, or the function is badly behaved. A good practice is to combine a robust bracket with a fast local method.

3. Kepler's equation as a worked example

For an elliptical orbit, the mean anomaly M and eccentric anomaly E satisfy

$$M = E - e \sin E. \tag{2}$$

This cannot be inverted with elementary functions. Newton's method gives

$$E_{n+1} = E_n - \frac{E_n - e \sin E_n - M}{1 - e \cos E_n}. \tag{3}$$

For modest eccentricity, $E_0 = M$ is often a good start. For high eccentricity and small M, convergence can be slower and a better initial guess or bracketing helps.

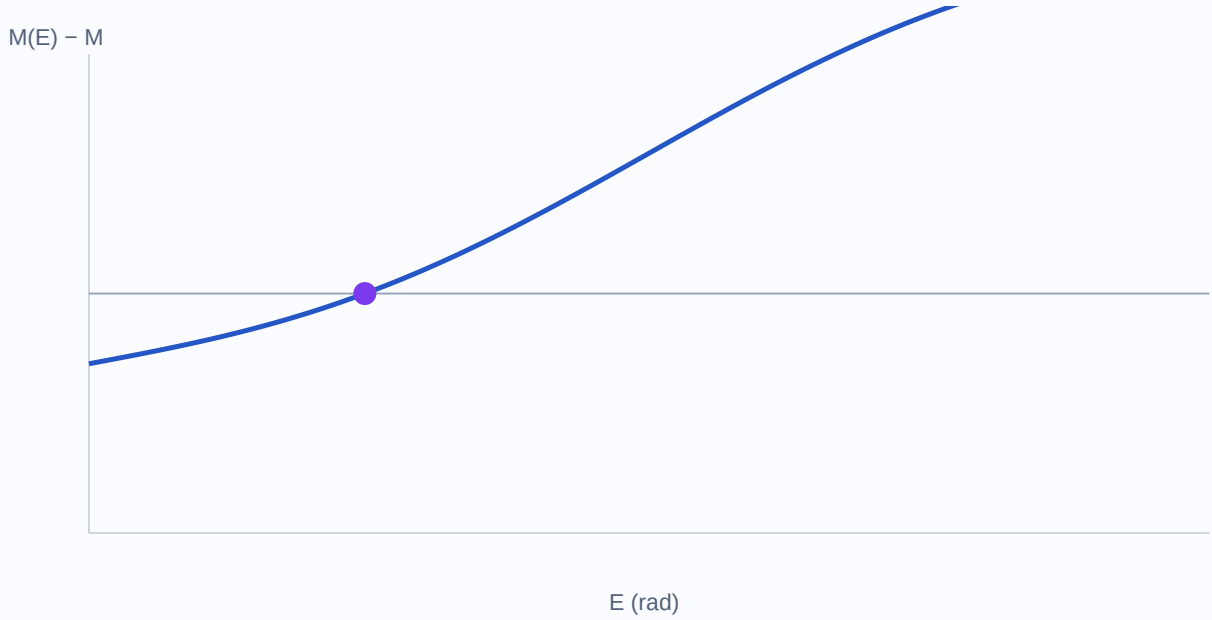
INTERACTIVE EXPERIMENT

Solve Kepler’s equation by iteration.

Solve M = E − e sin E for the eccentric anomaly E. Compare convergence speed and failure modes of three standard methods.

Eccentric anomaly E	Residual	Iterations used
88.6398°	**0.00e+0**	**5**

Change in last step
7.35e-14



Newton uses the local slope; bisection brackets a root; fixed-point iteration repeats E = M + e sin E. Convergence is not guaranteed for every guess.

Try this. Set e near 0.9 and M near 0. Which method struggles? Why is a good initial guess valuable?

Snapshot of the experiment. Open the HTML edition to change the controls.

4. Numerical differentiation and integration

A central difference approximates a derivative:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}. \tag{4}$$

If h is too large, truncation error dominates; if h is too small, round-off error dominates. There is usually an optimal intermediate step.

Numerical integration adds many small contributions. The trapezoidal rule uses straight-line segments; Composite Simpson's rule uses parabolic segments, requires an even number n of equal subintervals, and is usually more accurate for smooth functions:

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f_0 + 4f_1 + 2f_2 + \cdots + 4f_{n-1} + f_n\right]. \tag{5}$$

5. Differential equations and time stepping

Many physical laws are differential equations. The simplest time stepper is Euler's method:

$$y_{n+1} = y_n + h\,f(t_n, y_n). \tag{6}$$

It is easy but can be inaccurate or unstable. Runge–Kutta methods sample the slope several times per step and are much more reliable. In orbital mechanics, energy conservation and long-term stability often matter more than the error in a single step.

6. Error, convergence and sanity checks

A numerical answer should be accompanied by a residual, step-size test or convergence test. Halve the step size and see whether the answer changes by the expected amount. Check limiting cases: circular orbit, zero noise, constant function, or known analytic solution.

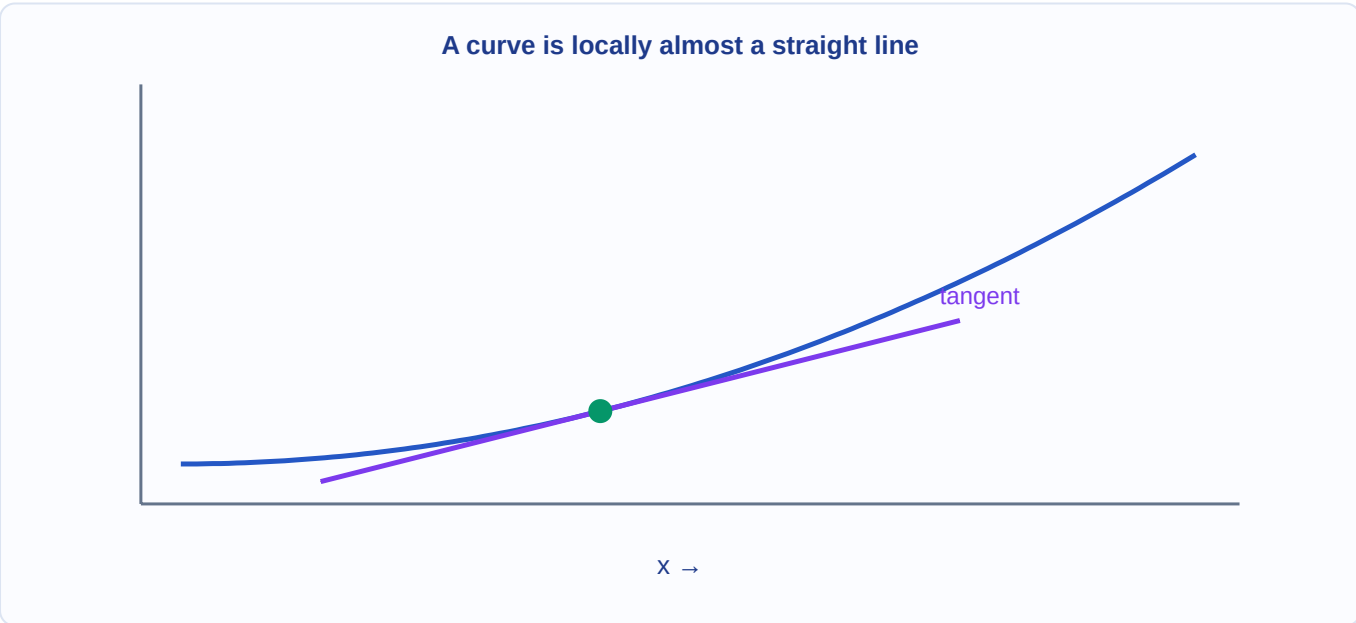


Figure 2. Small-angle and binomial approximations are the workhorses of Olympiad estimation.

A reliable method. State the equation, choose a method suited to its behaviour, iterate until a tolerance is met, and verify with a second method or limiting case. Never trust a number just because a computer printed it.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Bisection width. A root is bracketed in $[0, 2]$. How many bisection steps are needed to make the interval width less than 10^{-3}?

Worked solution →

[2] Problem 2 — One Newton step. Solve $x^2 - 2 = 0$ starting from $x_0 = 1.5$. Perform one Newton iteration and estimate the error.

Worked solution →

[2] Problem 3 — Kepler's equation. For $M = 1.0$ rad and $e = 0.5$, perform one Newton step from $E_0 = M$. Report the new E and the residual.

Worked solution →

[1] Problem 4 — Central difference. Use $h = 0.01$ to estimate the derivative of $f(x) = x^2$ at $x = 3$. Compare with the exact derivative.

Worked solution →

[2] Problem 5 — Trapezoidal rule. Estimate $\int_0^1 x^2\,dx$ with two equal trapezoids. Compare with the exact value and state whether the estimate is high or low.

Worked solution →

[2] Problem 6 — Simpson's rule. Estimate $\int_0^1 x^2\,dx$ with Simpson's rule using $n = 2$. Why is the result exact here?

Worked solution →

[2] Problem 7 — Euler decay. Use Euler's method with $h = 0.1$ to estimate $y(0.2)$ for $dy/dt = -y$, $y(0) = 1$. Compare with $e^{-0.2}$.

Worked solution →

[1] Problem 8 — Fixed-point iteration. For $E = M + e\sin E$ with $M = 0.5$ and $e = 0.2$, compute two fixed-point iterations starting from $E_0 = M$.

Worked solution →

[3] Problem 9 — Convergence order. A method has errors $10^{-1}, 10^{-2}, 10^{-4}, 10^{-8}$ on successive iterations. Is this consistent with linear or quadratic convergence? Why does Newton's method often behave this way near a simple root?

Worked solution →

[3] Problem 10 — Choose a method. You need the time of periapsis from a noisy light curve, and the model is expensive to evaluate. Would you use bisection, Newton's method, or a hybrid? Justify your choice and state one safeguard.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.